International Journal of Heat and Mass Transfer 170 (2021) 121003

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Numerical simulation of heat transfer and fluid flow during nanosecond pulsed laser processing of $Fe_{78}Si_9B_{13}$ amorphous alloys

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ARTICLE INFO

Article history:
Received 1 October 2020
Revised 28 December 2020
Accepted 16 January 2021
Available online 3 February 2021

Keywords:
Fe-based amorphous alloy
Nanosecond pulsed laser processing
Numerical simulation
Fluid flow
Heat transfer

ABSTRACT

Nanosecond pulsed laser processing is a promising method to modify the microstructure of amorphous alloy. However, the heat and mass transfer mechanisms during processing are still unclear, which prevent regulating the microstructure precisely. In this study, a 3D heat transfer and flow coupling model is established to investigate the heat transfer and fluid flow behaviors, and their effects on the surface morphology and microstructure formation during the nanosecond pulsed laser processing of $Fe_{78}Si_9B_{13}$ amorphous alloy. The physical processes such as melting and solidification, melt evaporation, melt convection and heat conduction, as well as the main driving forces such as surface tension, recoil pressure and gravity are taken into account in this model. The simulation morphology is in good agreement with the experimental results, indicating the validity of this numerical model. Based on simulation data analysis, it is found that the evolution mechanism of surface morphology is mainly related to the recoil pressure caused by evaporation and the Marangoni effect induced by the surface tension spatial gradient. Combined the temperature field analysis with experimental characterization, the formation mechanism of new amorphous phase in the molten zone (cooling rate at the order of 10^7 K/s) and the precipitation behavior of α-Fe(Si)/Fe-B phase nanocrystals in the heat-affected zone are clarified.

1. Introduction

The amorphous materials are significantly different from the corresponding crystalline alloys due to their disordered and metastable structure, rendering superior properties such as high strength, high hardness and excellent corrosion resistance [1,2]. Among various kinds of amorphous alloys, Fe-based amorphous alloys are used widely in power transformers and magnetic sensors, because of their excellent soft magnetic properties, outstanding corrosion resistance and ultrahigh strength [3,4].

To further increase the operational efficiency and service life of equipment such as power transformers, many researchers have made great efforts on improving the performance of Fe-based amorphous alloys. Fe-based amorphous alloys exhibit superior soft magnetic properties in their partially devitrified condition since it has been well established that nanocrystals embedded in an amorphous matrix lead to optimum properties [5–7]. In general, such partial devitrification can be achieved through furnace annealing the amorphous melt-spun ribbons for various temperature-time combinations [8–10]. Apart from the common furnace annealing for devitrification of amorphous, some recent studies have investigated devitrification via rapid annealing using specifically designed furnaces (heating rates of 100 - 200 K/s) [11] as well as dc Joule heating (heating rates of 50 - 200 K/s) [12]. Rapid heating allows nano-crystallization to occur in a much shorter time than in conventional anneals. As a consequence, grain nucleation is enhanced, grain growth is slowed, and the relatively slow processes leading to embrittlement are partially avoided. Laser processing is a most promising technique for devitrification of amorphous since the high energy density characteristic of the laser makes rapid heating and cooling possible (10^5-10^8 K/s) [13,14]. The distinct thermal histories of laser processing could influence the structure evolution in different zone. It have been reported that laser processed amorphous alloy exhibits significant structural heterogeneities: fully amorphous structure in molten pools, whereas a mixture of amorphous phase and nanocrystals in the heat affected zones [15–17]. The effect of distinct structural heterogeneities on the microstructure and properties of amorphous alloys are quite interesting. The earlier paper of our research group has studied the structural transformation and property improvement of $Fe_{78}Si_9B_{13}$ amorphous ribbon by nanosecond pulsed laser processing [18].

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https://doi.org/10.1016/j.ijheatmasstransfer.2021.121003
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In order to control the pulsed laser process to obtain a desired property improvement for amorphous alloy, researchers have tried to study the mechanism of laser-materials interaction by simulation method. The research of Chen [19] used the finite volume method to simulate the temperature distribution during laser surface melting for Zr-based amorphous alloy. A multi-physics model based three dimensional finite element (FE) heat transfer model on COMSOL™ platform was developed to predict the thermal histories generated during laser treatment of Fe-based metallic glass ribbons in Alam's paper [6]. The current researches focus on the temperature field simulation without considering the influence of flow field on temperature evolution and structure formation. In addition, many numerical heat transfer and fluid flow models of laser process have been reported to study the transient nature of fluid flow, heat transfer and solidification characteristics based on CFD (Computational Fluid Dynamics) software Fluent. But they were common used in laser welding processing to study the keyhole dynamic behavior and the formation of pores/voids [20–23]. However, there is little research about numerical simulation of temperature field and fluid flow for pulsed laser processing of amorphous alloys. In the very short time of nanosecond laser pulsed processing, how is the transient flow field and temperature field distributed in the area of action (molten pool and heat affected area) and how does it affect the formation of surface morphology and microstructure deserve further study.

In this study, a three dimensional numerical model was developed to analyze the heat transfer and fluid flow behaviors, and their effects on the surface morphology and microstructure formation of $Fe_{78}Si_9B_{13}$ amorphous alloys during the nanosecond pulsed laser processing. The rest of this paper is organized as follows. First, the experimental method is shown in Section 2. Then, the numerical model, including basic assumption, governing equation, surface tracking method, boundary condition and etc., is described in Section 3. In Section 4, the transient flow field and temperature field are analyzed to explain the evolution of surface morphology and microstructure. Finally, main conclusions are given.

2. Experimental methods

The amorphous ribbons with a nominal composition of $Fe_{78}Si_9B_{13}$ were prepared by a single-roller melt-spinning technique, and they were supplied by Qingdao Yunlu Energy Technology Company, Qingdao, China. Commercial ribbons with 5 cm in width and 30 μm in thickness were cut to a length of 10 cm for nanosecond pulsed laser processing. The ribbons were processed using the Ytterbium MOPA pulsed fiber laser (TieDan-B50). During the nanosecond pulsed laser processing, the parameters were set as constant values except the laser power. The scan speed is 3000 mm/s, the pulsed width is 200 ns, the frequency is 20 kHz and the tracks spacing is 0.02 mm. The laser power (P_L) was set at 10, 20 and 30 W, respectively. The type of scanning galvanometer was SACNcube III 14. The surface morphology of samples was examined by scanning electron microscopy (SEM, SUPRA55, Zeiss, Germany). The SEM sample was 5 × 5 mm square which was cutted from the laser processed ribbons, and the surface was cleaned with alcohol. Transmission electron microscopy (TEM, JEM-2100) was used to further observe the microstructure of samples, and the Fourier filtering for HRTEM images was using DigitalMicrograph software. The amorphous ribbons (30 μm in thickness) were cutted into 3 mm round piece firstly, and then pre-thinned using a single jet electrolytic with 4% perchloric acid and 96% alcohol electrolyte at −20°C until a hole was formed. Lastly, argon ion thinning was used to obtain the suitable sample for TEM test.

3. Numerical model

3.1. Physical description of nanosecond pulsed laser processing

During the nanosecond pulsed laser processing, most of the laser energy is absorbed by the surface layer of the material, which results in a significant rise of temperature and forms a small molten pool. On the other hand, a slight volume of melt on the surface layer can vaporize quickly as the temperature exceeds the boiling temperature. The general heat transfer mechanism includes the heat radiation, heat conduction and heat convection. The three coupled heat transfer mechanisms cause the thermal behavior during nanosecond pulsed laser processing to become very complex. Fig. 1 shows the schematic of the physical model for nanosecond pulsed laser processing. In addition, the mass transfer caused by recoil pressure, surface tension, hydrostatic pressure and hydrodynamic pressure are all needed to be considered in the numerical model.

3.2. Model development and basic assumption

In order to make the model tractable, the following assumptions are proposed. During nanosecond pulsed laser processing, the laser energy can be regarded as surface heat sources, which obeys the Gaussian distribution. The $Fe_{78}Si_9B_{13}$ amorphous alloy is considered as homogeneous material. Moreover, the fluid motion in the molten pool is assumed to be Newtonian, laminar and incompressible. The properties parameters of $Fe_{78}Si_9B_{13}$ amorphous alloys list in Table 1.

Fig. 2 illustrates the schematic of the computational domain. The computational domain was set as 120 μm in length, 60 μm in width and 40 μm in depth with 10 μm height of air domain to track the free surface of molten pool induced by nanosecond pulsed laser processing. In order to obtain more details information about the morphology evolution and temperature distribution during the nanosecond pulsed laser processing, the middle fluid flow zone use the fine mesh with size as 0.25 μm and a relatively

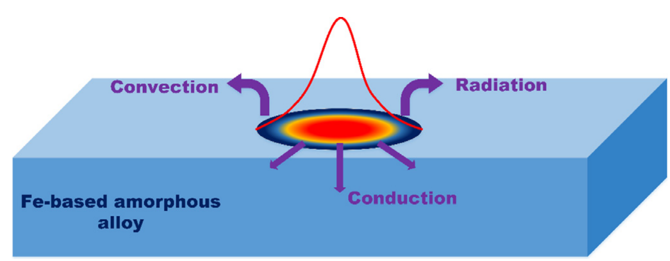

Fig. 1. Schematic of the physical model for nanosecond pulsed laser processing.

Table 1
Properties parameters of $Fe_{78}Si_9B_{13}$ amorphous alloys [6,36].

Physical properties	Value
Density (kg/m^3)	7180
Solid specific heat (J/kg• K)	500
Liquid specific heat (J/kg• K)	900
Thermal conductivity (W/m•K)	9 T < 1455 K 0.015T+5 T >1455 K
Liquidus temperature (K)	1455
Solidus temperature (K)	1373
Boiling temperature (K)	3100
Crystallization temperature (K)	799
Emissivity	0.7
Latent heat of fusion (J/kg)	2.20e+5
Latent heat of vaporization (J/kg)	6.52e+6
Surface tension coefficient (N/m)	γ=1.0 − 0.0004(T − 1455)

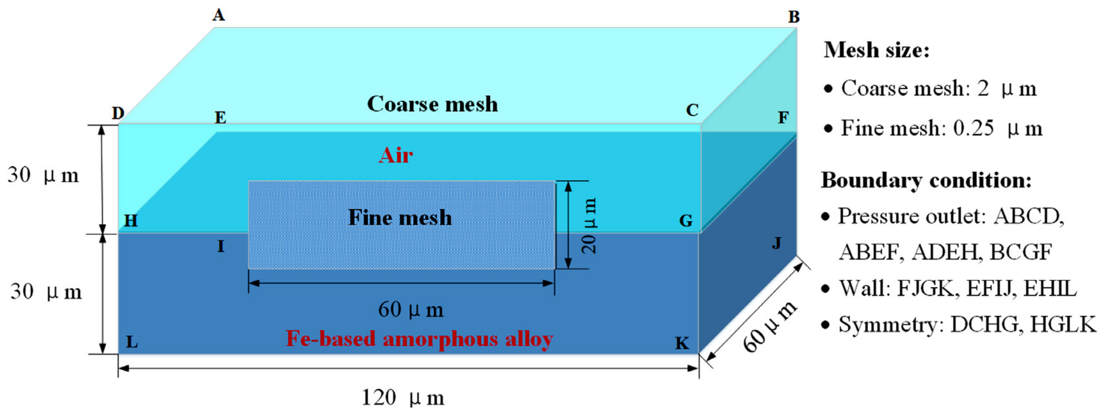

Fig. 2. Schematic of the computational domain.

coarse mesh (2 μm) is adopted for the remaining zone. The front domain surface is set as symmetric to reduce the computational cost, which still result in an excessive number of grids (3645000). So, only one pulse process is considered in the simulation. Because the laser blob with coronal-like morphology in the experiment results was caused by only one pulse, the single pulse process in the simulation can fully explain the mechanism of surface morphology evolution. In addition, the study of a single pulse process of the temperature distribution is the basis and can provide the necessary information for the structural transformation of Fe-based amorphous alloy during nanosecond pulsed laser processing.

3.3. Governing equations

The governing equations describe the heat and mass transfer phenomena and fluid dynamics for nanosecond pulsed laser processing, including mass, momentum and energy conservation equations [24,25].

Continuity equation:

$$\nabla \cdot V = 0 \tag{1}$$

where V is the velocity vector of fluid flow.

Momentum equation:

$$\rho\left(\frac{\partial V}{\partial t} + V \cdot \nabla V\right) = -\nabla P + \mu \nabla^2 V - C_{\text{mushy}} \frac{(1-f_l)^2}{f^3 + A} V + \rho g \tag{2}$$

where ρ is fluid density, t is time, P is pressure, C_{mushy} is the mushy zone constant, A is a small positive number (To prevent the denominator being zero during the calculation). The third term on the right-hand of Eq. (2) is the momentum sink due to the reduced porosity in the mushy zone. g is gravitational acceleration. μ is the dynamic viscosity, f_l is the liquid fraction, which is defined as:

$$f_l = \begin{cases} 0 & T < T_s \\ \frac{T-T_s}{T_l-T_s} & T_s \le T \le T_l \\ 1 & T > T_l \end{cases} \tag{3}$$

where T_s is the solidus temperature and T_l is the liquidus temperature.

Energy equation:

$$\rho\left(\frac{\partial h}{\partial t} + V \cdot \nabla h\right) = \nabla \cdot (\lambda \nabla T) \tag{4}$$

The energy is formulated by enthalpy continuum model in the solid-liquid phase transition:

$$h = \begin{cases} \rho C_{ps} T & T < T_s \\ h(T_s) + f_l \times L_f & T_s \le T \le T_l \\ h(T_l) + \rho C_{pl}(T - T_l) & T > T_l \end{cases} \tag{5}$$

where h is the enthalpy of material, λ is the thermal conductivity, C_{ps} is the specific heat of solid phase and C_{pl} is the specific heat of liquid phase, L_f is the latent heat of fusion, f_l is the liquid fraction as described by Eq. (3).

3.4. VOF model

The VOF model can model two or more immiscible fluids by solving a single set of momentum equations and tracking the volume fraction of each fluid throughout the domain.

The tracking of the interface(s) between the phases is accomplished by the solution of a continuity equation for the volume fraction of one (or more) of the phases. The volume fraction F is governed by the following conservation equation [25,26]:

$$\frac{\partial F}{\partial t} + \nabla \cdot \left(\vec{V} F\right) = 0 \tag{6}$$

where F is the volume fraction of phase, $F = 1$ when a cell is full of liquid, whereas $F = 0$ when a cell is full of gas. A cell with $0 < F < 1$ is on the free surface of liquid and gas.

3.5. Heating source model

The Gaussian heat flux input is used on the top surface. The heat distribution is expressed as follows [27,28]:

$$q(r) = \frac{3\eta Q}{\pi r^2} \exp\left(\frac{-3\left(x^2 + y^2\right)}{r^2}\right) \tag{7}$$

here, r is the radius of Gauss surface, η is energy transfer efficiency, Q is the laser energy.

3.6. Boundary condition

The external surfaces of air phase domain are set as pressure outlet. For the Fe-based amorphous alloy phase domain, the bottom and lateral planes are set as wall boundaries with heat loss by convection and radiation. The top surface of Fe-based amorphous alloy is treated as free surface tracked by VOF method, and the laser heat and pressure act on that surface as source terms.

For the top surface of $Fe_{78}Si_9B_{13}$ amorphous alloy phase domain, the thermal boundary condition can be expressed as follows:

$$-\lambda \frac{\partial T}{\partial n} = q_{\text{laser}} - q_{\text{cov}} - q_{\text{rad}} - q_{\text{evp}} \tag{8}$$

where λ is thermal conductivity, T is temperature, n is the magnitude of normal vector of local surface, q_{laser} is the laser heat input, q_{cov} is heat loss by convection, q_{rad} is heat loss by radiation and

q_{evp} is heat loss by evaporation, which can be expressed as follows [21,24,25].

$$q_{cov} = h_c(T - T_0) \tag{9}$$

$$q_{rad} = \sigma_{SB}\varepsilon\left(T^4 - T_0^4\right) \tag{10}$$

$$q_{evp} = 0.82\frac{\Delta H^*}{\sqrt{2\pi M R_{gu} T}} P_0 \exp\left(\Delta H^* \frac{T - T_{LV}}{R_{gu} T T_{LV}}\right) \tag{11}$$

$$\Delta H^* = \Delta H_{LV} + \frac{\gamma_C(\gamma_C + 1)}{2(\gamma_C - 1)} R_{gu} T \tag{12}$$

Here h_c is the convective heat transfer coefficient, T_0 is the ambient temperature, σ_{SB} is the Stephan-Boltzmann constant, ε is the material emissivity, ΔH^* is enthalpy of vapor flowing at sonic velocity, M is atomic mass, R_{gu} is the universal gas constant, T_{LV} is the equilibrium temperature between gas phase and liquid phase, ΔH_{LV} is the latent heat of evaporation and γ_c is a proportionality factor. Moreover, it is difficult to detect the transient heat transfer coefficient and material emissivity of the $Fe_{78}Si_9B_{13}$ amorphous alloys, and the temperature dependent data is lacked in the literature. Therefore, several simplifications (set as constant) may be used in simulations, and the reported simulated results agree well with the experimental results in general (for example, laser welding and additive manufacturing) [29–32]. In this work, we also assume the emissivity and heat transfer coefficient as a constant.

On the wall boundary, there is no heat input but still have the heat loss by convection and radiation.

$$-\lambda\frac{\partial T}{\partial n} = -q_{cov} - q_{rad} \tag{13}$$

For the nanosecond pulsed laser processed surface, several kinds of pressures are considered: recoil pressure, the surface tension, radiation pressure, hydrostatic pressure and hydrodynamic pressure. Since the value of radiation pressure is quite small compared with the others, it is omitted in the model. The pressure equilibrium on the surface should follow the equation:

$$P = P_r + P_\sigma + P_s + P_d \tag{14}$$

Here, P_r is the recoil pressure, which can be expressed as follows [21,33]:

$$P_r = 0.54P_0 \exp\left(\Delta H_{LV}\frac{T - T_b}{RTT_b}\right) \tag{15}$$

Here, P_0 is the atmospheric pressure, ΔH_{LV} is the latent heat of evaporation, T_b is the liquid-vapor equilibrium temperature, T is the surface temperature and R is the universal gas constant.

The surface tension P_σ can be given by [34]

$$P_\sigma = K\gamma \tag{16}$$

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_0) \tag{17}$$

$$K = -\left[\nabla \cdot \left(\frac{\vec{n}}{|\vec{n}|}\right)\right] = \frac{1}{|\vec{n}|}\left[\left(\frac{\vec{n}}{|\vec{n}|} \cdot \nabla\right)|\vec{n}| - \left(\nabla \cdot \vec{n}\right)\right] \tag{18}$$

where K is free surface curvature and γ is the surface tension coefficient.

The hydrostatic pressure P_s and hydrodynamic pressure P_d are given as follows [35]:

$$P_s = gh \tag{19}$$

$$P_d = \frac{\upsilon^2}{2g} \tag{20}$$

here g is the gravitational acceleration, h is the height of the fluid and v is the velocity of fluid.

3.7. Numerical method

The commercial CFD software Fluent was used to develop the model, and a UDF (User-Defined Function) file written by C language was used to realize the laser source term and boundary condition. The mass, momentum and energy conservation equations were discretized using control volume method and then solved using the Pressure-Implicit with Splitting of Operators

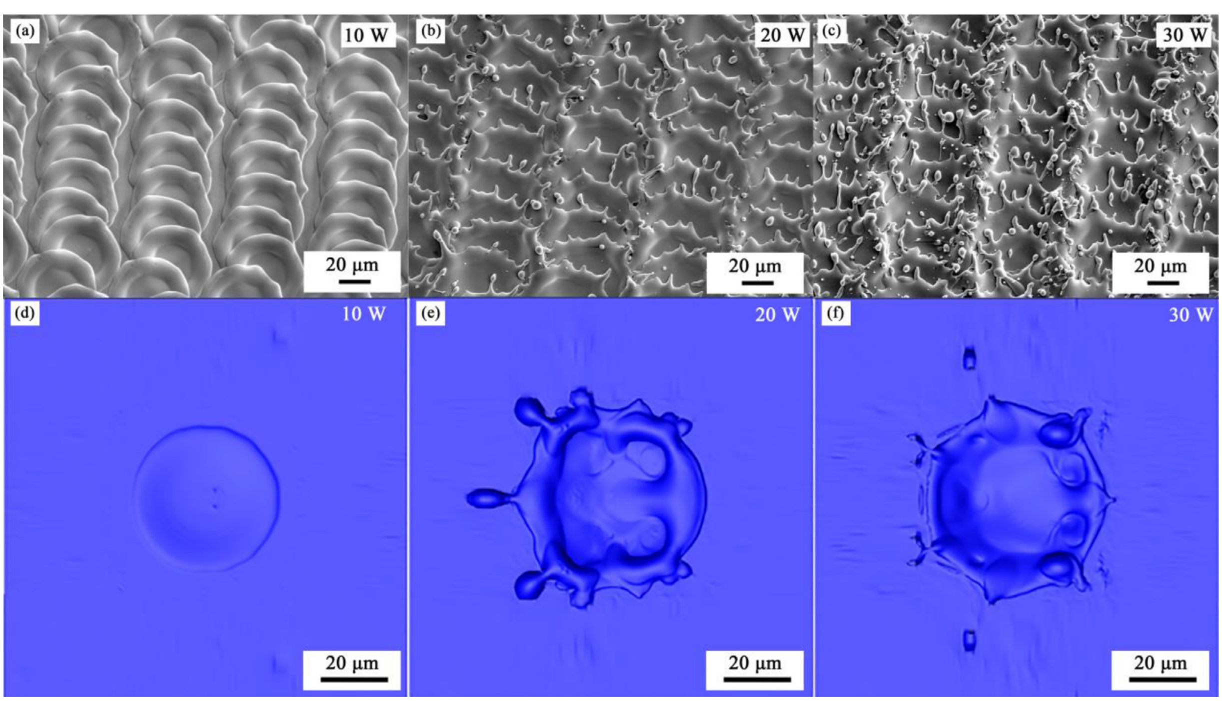

Fig. 3. Comparison of experimental and simulated surface morphology of $Fe_{78}Si_9B_{13}$ amorphous alloys after nanosecond pulsed laser processing.

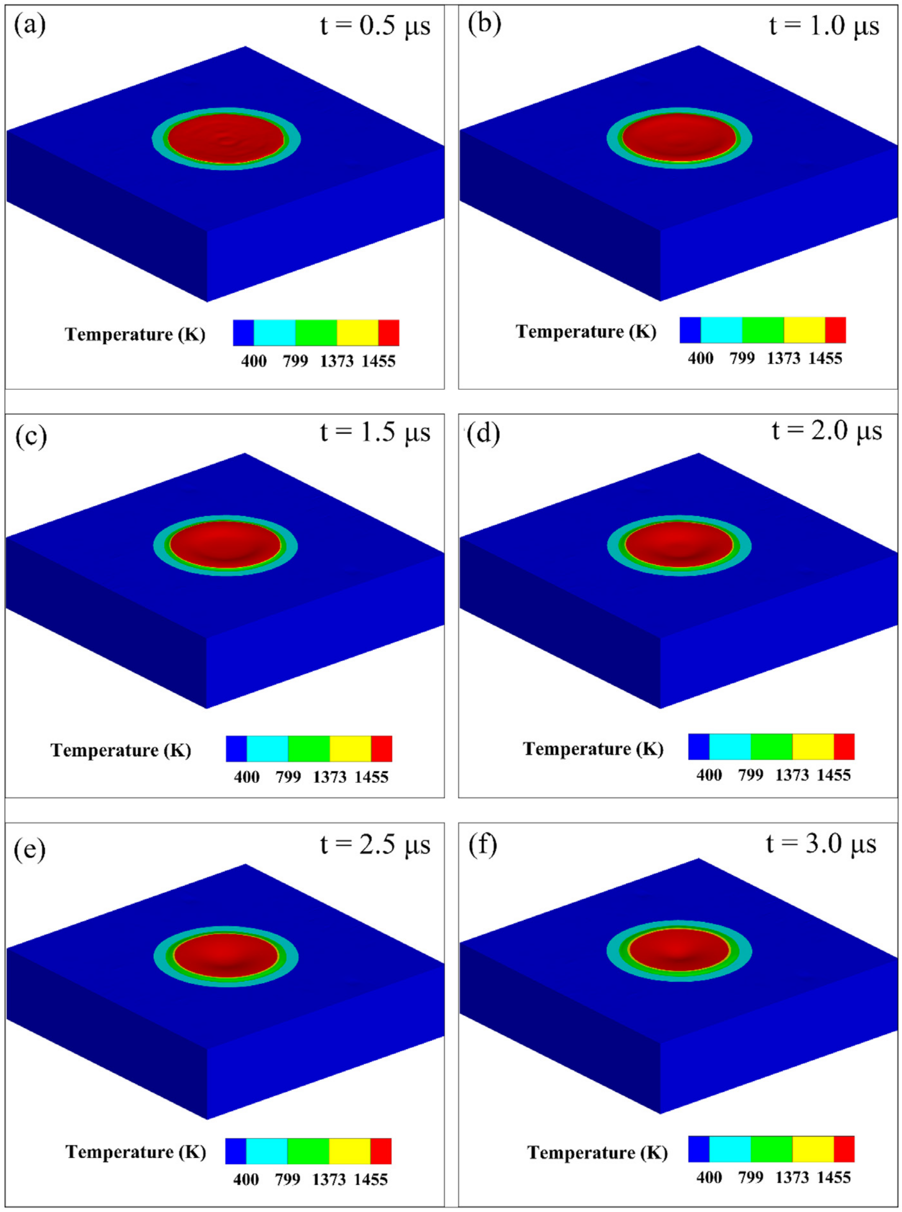

Fig. 4. The surface morphology evolution of $Fe_{78}Si_9B_{13}$ amorphous alloys with the pulsed laser power as 10 W.

(PISO) pressure-velocity coupling algorithm. The time step was restricted to be no more than 5×10^{-9} s to satisfy the convergence criteria requested by the governing equations, as well as the movement of the liquid/gas interface.

4. Results and discussion

4.1. Validation of the model

The numerical model was validated by comparing the simulation data with the corresponding experimental results. Fig. 3 compares the experimental and simulated surface morphologies of $Fe_{78}Si_9B_{13}$ amorphous alloy after nanosecond pulsed laser processing. The experimental results in Fig. 3(a)–(c) are the SEM images for samples after nanosecond pulsed laser processing with $P_L = 10$, 20 and 30 W, respectively. After nanosecond pulsed laser processing, the surface of $Fe_{78}Si_9B_{13}$ amorphous ribbons has obvious laser processed blobs, which are convex in perimeter and sunk in the middle, like coronal morphology. When increase P_L from 10 to 30 W, the spreading area of blobs increases and their shape becomes irregular with some splashes around the marginal of laser blobs. The simulation results with different laser power (P_L) show the same trend as the experimental results. For $P_L = 10$ W (Fig. 3(d)), the sample surface has a coronal-like morphology, which convexs on all sides and concaves in the middle. With increasing P_L (Fig. 3(e) and (f)), obvious splash phenomenon occurs on the sample surface, and the splash degree with sample with $P_L = 30$ W is higher than sample with $P_L = 20$ W. Due to the limited size of this model, the melt that flies out of the model is not considered, so the simulation results of surface morphology of samples with $P_L = 20$ and 30 W shows less splash droplets compared with experimental results. In general, the simulation results have a high degree of agreement with the experimental results, which verifies the effectiveness of this model.

4.2. Morphologic evolution analysis

This section is mainly about the flow field analysis of $Fe_{78}Si_9B_{13}$ amorphous alloy during nanosecond pulsed laser processing. Combined with the velocity field, pressure field and temperature field,

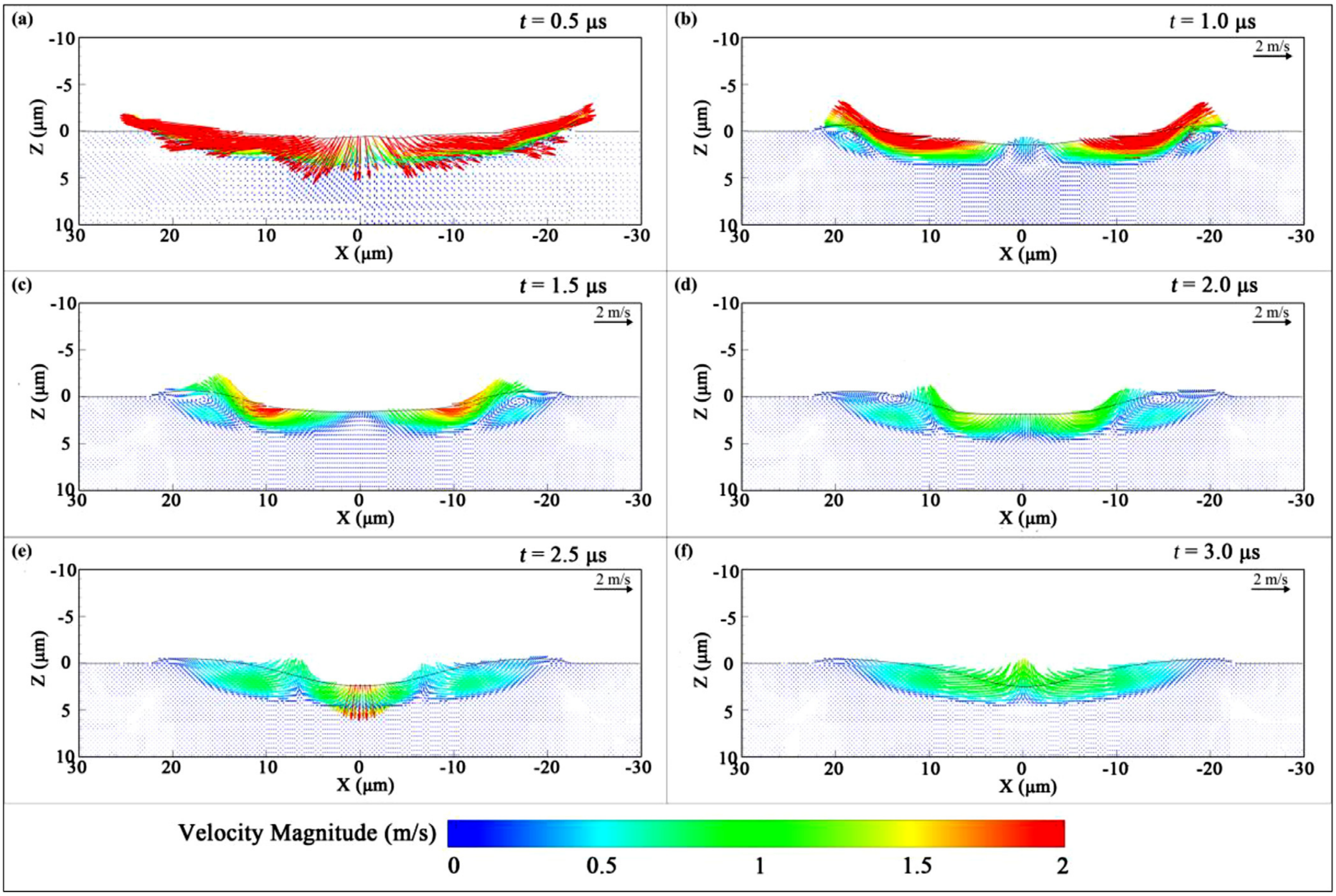

Fig. 5. Fluid flow field on section $Y = 0$ of $Fe_{78}Si_9B_{13}$ amorphous alloys with the pulsed laser power as 10 W.

the influence of nanosecond pulsed laser processing on the surface morphology evolution of $Fe_{78}Si_9B_{13}$ amorphous alloy is discussed as following.

The surface morphology evolution and temperature distribution of the $Fe_{78}Si_9B_{13}$ amorphous alloy with $P_L = 10$ W are displayed in Fig. 4. The selected time (t) values in the figure are 0.5, 1.0, 1.5, 2.0, 2.5 and 3.0 μs. The color of the figure shows the temperature distribution, and the contour level selects the crystallization temperature T_c, solidus temperature T_s and liquidus temperature T_l. At $t = 0.5$ μs, the surface temperature exceeds T_l, forming a small molten pool. The melt surface is in an unstable fluctuating state (Fig. 4(a)). At $t = 1.0$ μs, the surface molten pool tends to depress in the middle, as shown in Fig. 4(b). At $t = 1.5$ μs, the central depression expands and a small bulge forms around the pool, as shown in Fig. 4(c). However, the area of the central depression decreases at $t = 2.0$ μs (Fig. 4(d)). After this, the central depression area continues to decreases continually, having a tendency to recover, as shown in Fig. 4(e) and (f).

Fig. 5 shows the flow field of the $Fe_{78}Si_9B_{13}$ amorphous alloy with $P_L = 10$ W on the section $Y = 0$ with t values corresponding to Fig. 4. The color in the figure stands for the magnitude of the velocity (v). The v of the molten pool surface is relatively large at the beginning, but decreases with increasing t. By contrast, the v inside the molten pool increases with increasing t. During the nanosecond pulsed laser processing, the surface temperature of $Fe_{78}Si_9B_{13}$ amorphous alloys increases rapidly to reach the boiling point and causes evaporation. Severe recoil pressure induced by evaporation acts on the residual molten pool surface, which can overcome the static pressure and surface tension of molten pool itself. It can promote the depression of molten pool and make molten pool to expand. At $t = 0.5$ μs, the v of the molten pool surface is towards the interior of the molten pool, and its value is greater than 2 m/s, as shown in Fig. 5(a). In addition, the interaction of various forces causes the oscillation phenomenon in the molten pool, which corresponds to Fig. 4(a). At $t = 1.0$ μs, the effect of recoil pressure is apparently reduced. Simultaneously, the effects of surface tension and gravity increase. The Marangoni effect induced by the surface tension spatial gradient causes two symmetric vortexes at the molten pool edge (Fig. 5(b)). The vortexes cause the melt flows towards pool internal from the pool upper surface, which accelerates the heat transfer inside the molten pool. At $t = 1.5$ μs, the vortex region is further enlarged and the v of the molten pool surface begins to decrease due to the friction dissipation in the fluid (Fig. 5(c)). At $t = 2.0$ μs, the v of the molten pool surface decreases further, and the vortex caused by the Marangoni effect slowly advances towards the pool center and causes the melt in the pool to move upward from the bottom (Fig. 5(d)). At this time, the downward trend of the molten pool surface is restrained to a certain extent, and the molten pool depression tends to rise upward. At $t = 2.5$ μs, the surface velocity of the pool is further reduced and the vortex region is further enlarged (Fig. 5(e)). At $t = 3.0$ μs, the melt flows upward from the interior, indicating a recovery trend of the depressed area as shown in Fig. 5(f). Based on the above analysis, it can be concluded that the coronal-like morphology is mainly related to the recoil pressure, surface tension and gravity. At the beginning, the pool center has a distinct depression tendency under the impact of recoil pressure. With increasing t, the effects of surface tension and gravity are gradually increased and can cushion the impact of recoil pressure. The effect of recoil pressure decreases in cooling, and the depressed region tends to recover under

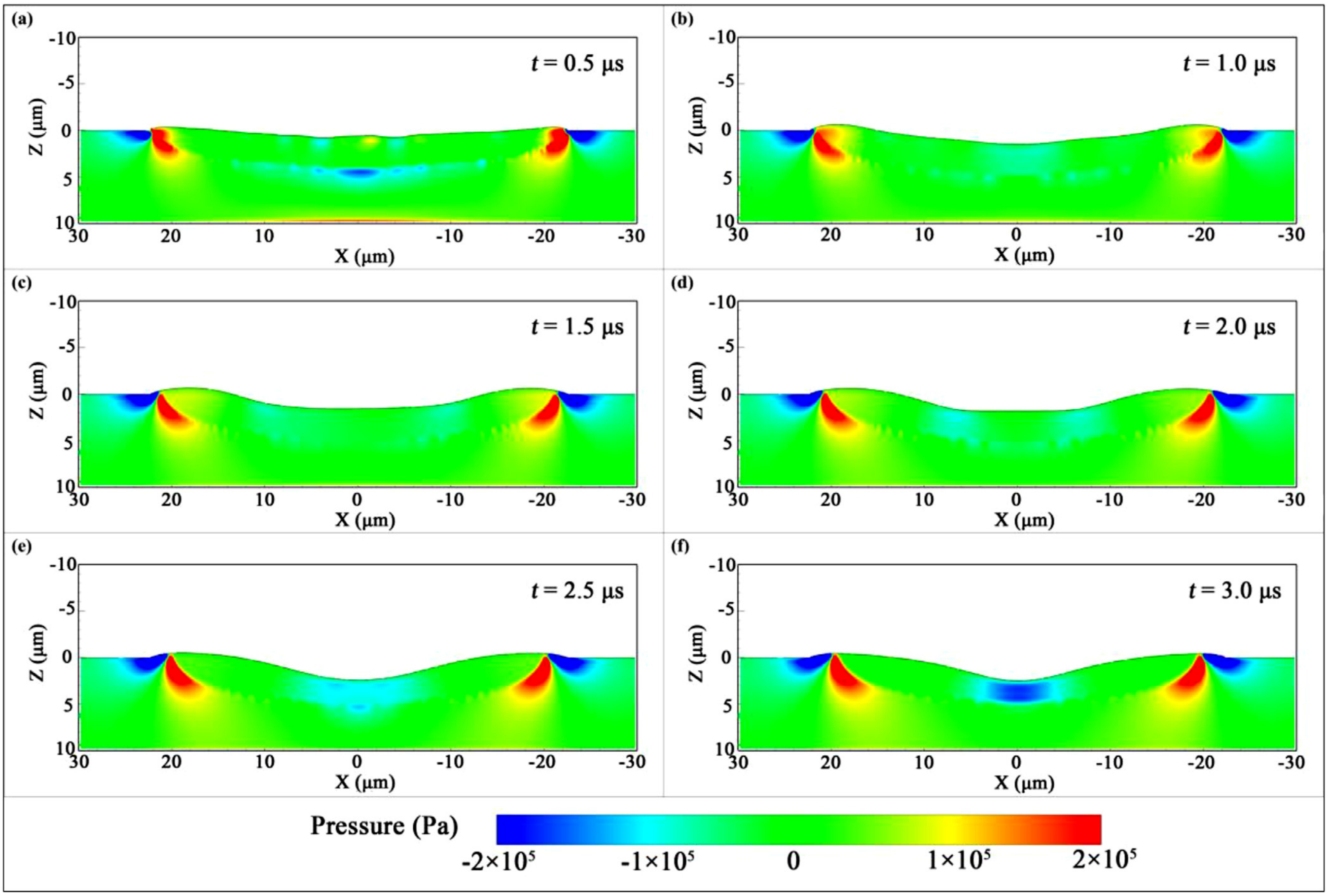

Fig. 6. Pressure profile on section $Y = 0$ of $Fe_{78}Si_9B_{13}$ amorphous alloys with the pulsed laser power as 10 W.

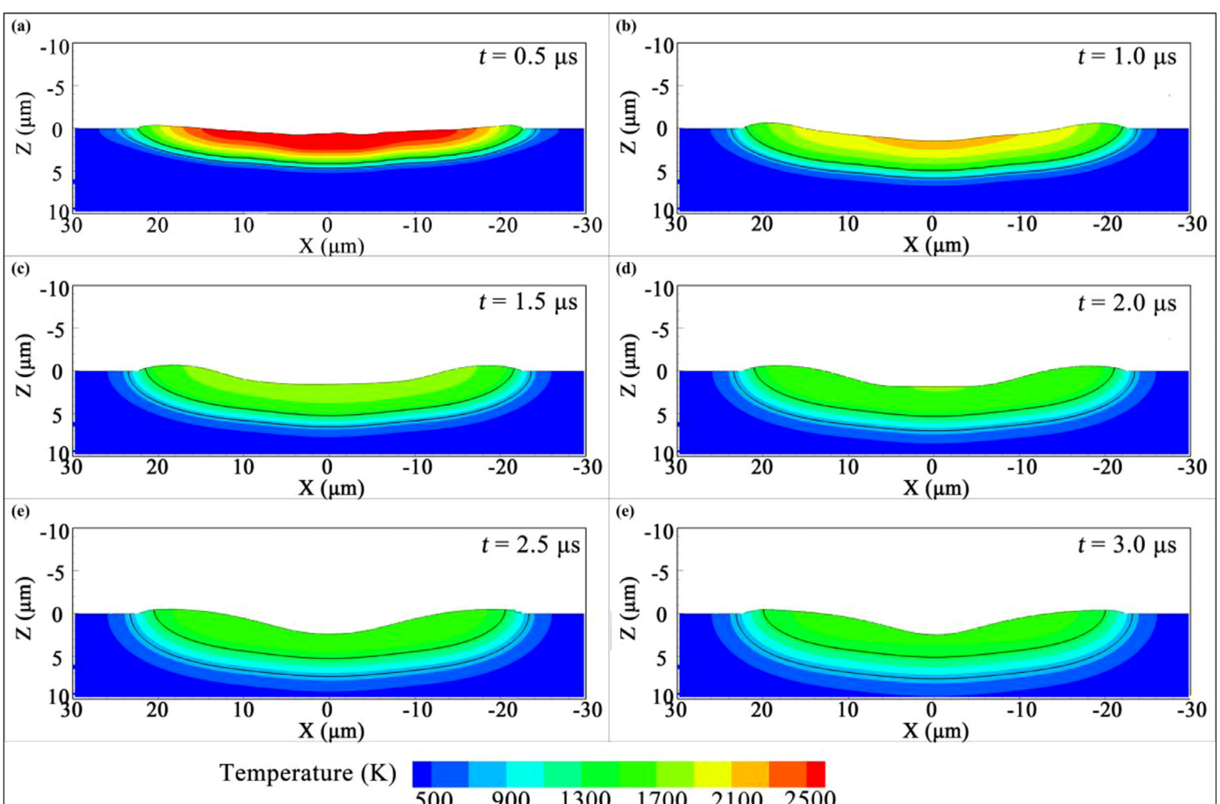

Fig. 7. Temperature distribution on section $Y = 0$ of $Fe_{78}Si_9B_{13}$ amorphous alloys with the pulsed laser power as 10 W.

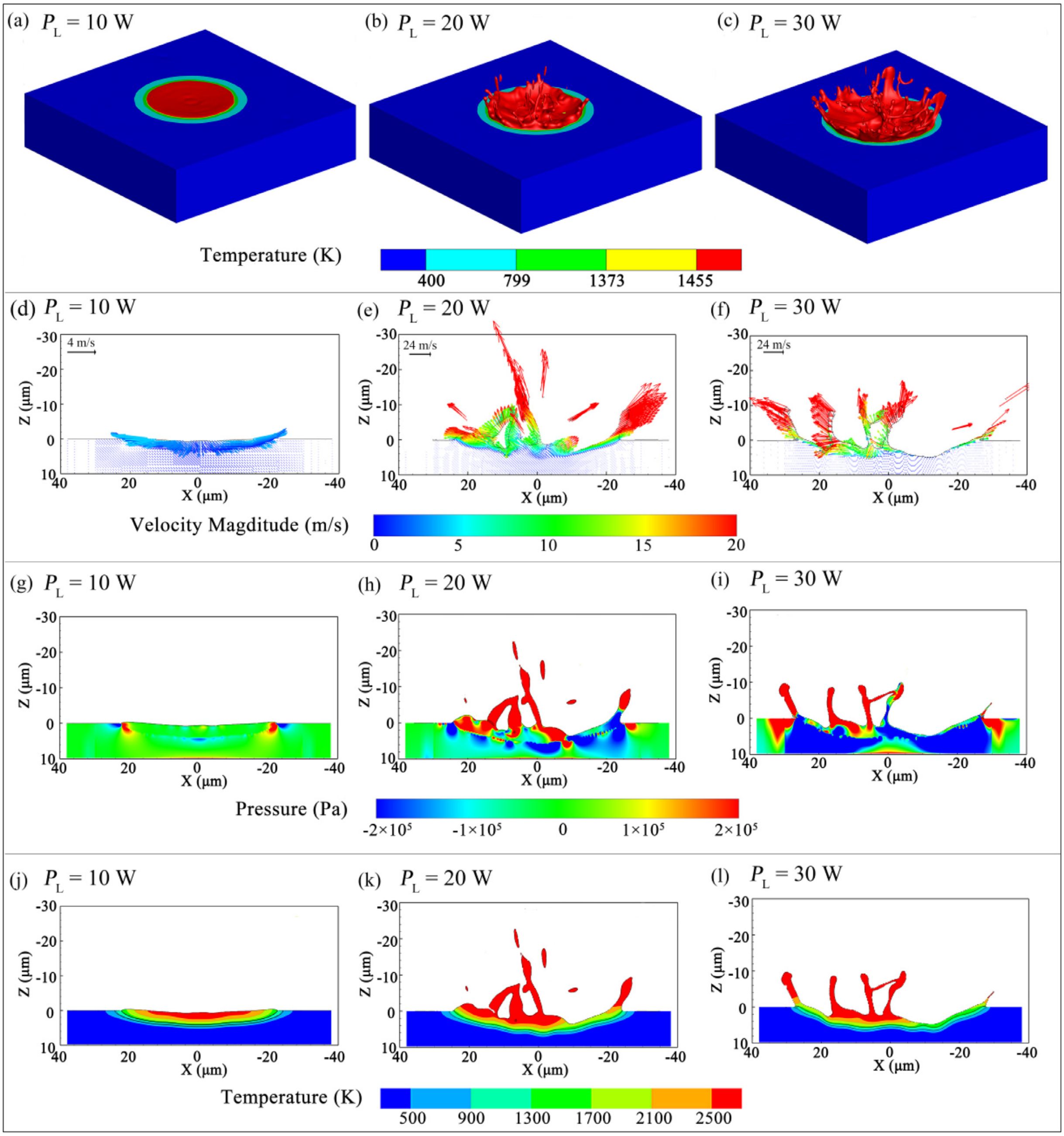

Fig. 8. The surface morphology, fluid flow field, pressure profile and temperature distribution of $Fe_{78}Si_9B_{13}$ amorphous alloys after nanosecond pulsed laser processing at 0.5 μs. (a), (d), (g) and (j): sample with P_L = 10 W; (b), (e), (h) and (k): sample with P_L = 20 W; (c), (f), (i) and (l): sample with P_L = 30 W.

the surface tension and gravity. In addition, the Marangoni effect induced by the surface tension spatial gradient also plays a important role in the formation of coronal-like morphology.

The pressure field distribution of the $Fe_{78}Si_9B_{13}$ amorphous alloy with $P_L = 10$ W on the section $Y = 0$ is shown in Fig. 6. The t values is also consistent with the morphological evolution figure in Fig. 4, and the color stands for the relative pressure (atmospheric pressure is taken as the reference). At $t = 0.5$ μs, a low pressure region (less than -200000 Pa) is near the solid-liquid interface line of the molten pool (Fig. 6(a)). With the flow of melt, the low pressure region at the bottom of the pool diffuses at $t = 1.0$ μs, creating a small area of low pressure (around -100000 Pa) in the pool center, as shown in Fig. 6(b). This low pressure region causes a slight flow trend of the melt in the intermediate region, slightly echoing the upward velocity flow field in the intermediate region of the molten pool in Fig. 5(b). At $t = 1.5$ and 2.0 μs, the low pressure region diffuses from the center to the sides until it moves to the edge of the vortex (Fig. 5(c) and (d)). At $t = 2.5$ μs, two areas of low pressure region merge slowly in the pool center due to the enlarge of vortex region (Fig. 6(e)), resulting in a transient increase of downward velocity for the melt in the center surface as shown in Fig. 5(e). At $t = 3.0$ μs, there is a low pressure region (pressure less than -200000 Pa) in the pool center (Fig. 6(f)). Combined with the flow field in Fig. 5(f), there is only a upward velocity in the molten pool, and the velocity in the middle of low pressure region is the maximum. It can be clearly seen that the pressure is concentrated on the edge of the upper surface of the molten pool, indicating a strong fluid flow trend here.

Fig. 7 shows the temperature field distribution of the $Fe_{78}Si_9B_{13}$ amorphous alloy with $P_L = 10$ W on the section $Y = 0$, and the t

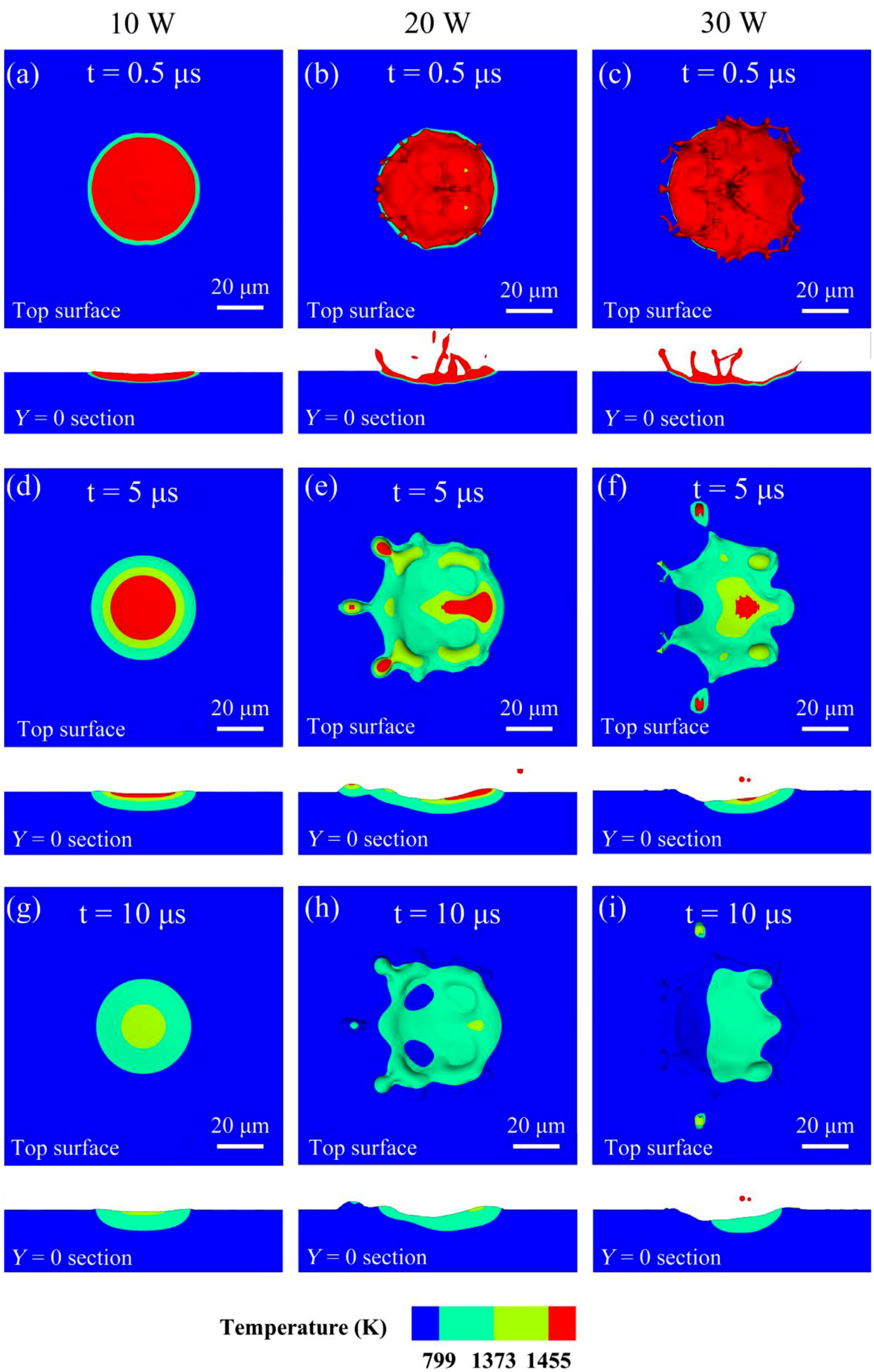

Fig. 9. The temperature distribution diagram of $Fe_{78}Si_9B_{13}$ amorphous alloy with different laser power and different time on the top surface and $Y = 0$ section. (a), (d) and (g): sample with $P_L = 10$ W; (b), (e) and (h): sample with $P_L = 20$ W; (c), (f) and (i): sample with $P_L = 30$ W.

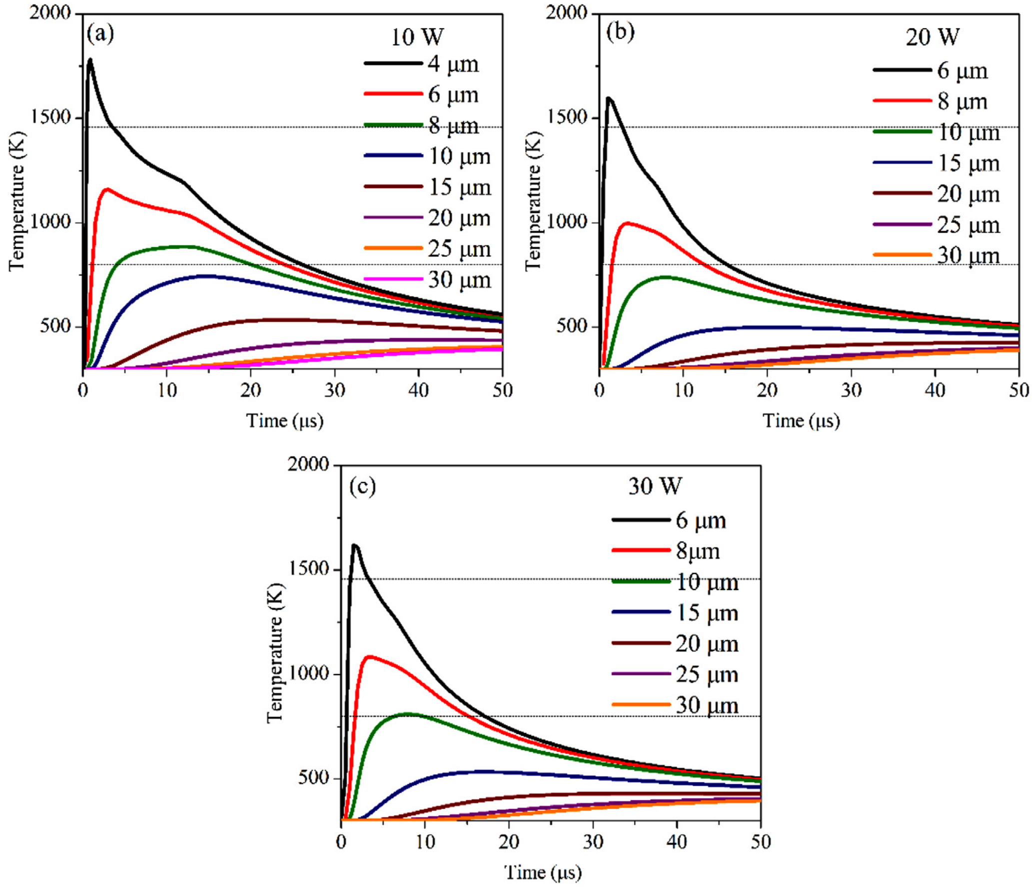

Fig. 10. The temperature curves of $Fe_{78}Si_9B_{13}$ amorphous alloy model at various Z during nanosecond pulsed laser processing. (a), (b), and (c): P_L =10, 20 and 30W.

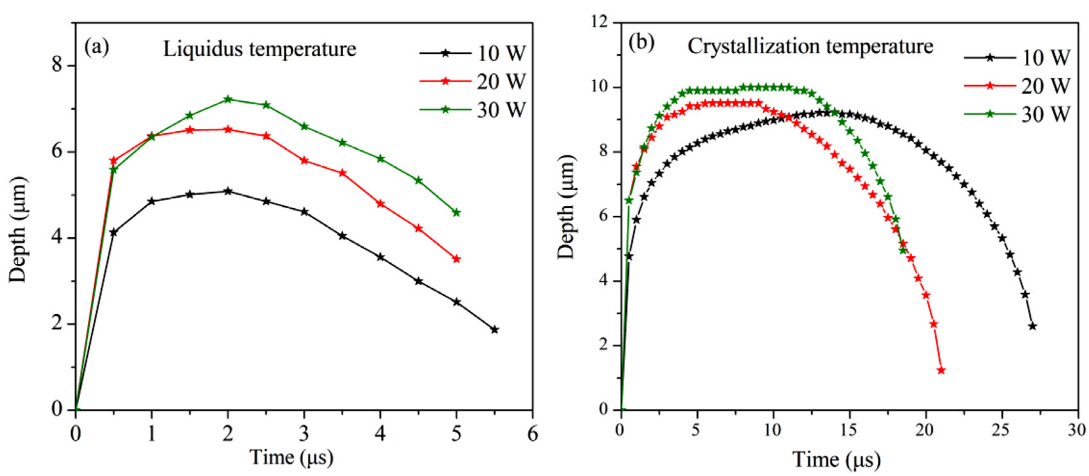

Fig. 11. Time variation of the maximum depths (Z direction) for (a) liquidus temperature and (b) crystallization temperature during nanosecond pulsed laser processing of $Fe_{78}Si_9B_{13}$ amorphous alloys.

values is also consistent with the morphological evolution figure in Fig. 3. The color bar in the figure shows the temperature distribution, and two black lines denotes T_s and the T_c position, respectively. There are obvious temperature gradients in the radial direction (Y direction) and longitudinal direction (Z direction) of the laser acting region (Fig. 7(a)), which is the root reason of the Marangoni effect. After nanosecond pulsed laser processing, the surface temperature of amorphous alloy increases rapidly, and then the heat is dissipated by heat radiation, convection and conduction. In addition, the vortex region caused by the Marangoni effect transfers the heat from the upper part to the inner part of the molten pool, which promotes the heat dissipation process on the surface of the molten pool (Fig. 7(b)–(f)). Moreover, the maximum depth of the molten pool is about 5 μm and does not increase at 3.0 μs (Fig. 7(f)).

The surface morphology, fluid flow field, pressure profile and temperature distribution of the $Fe_{78}Si_9B_{13}$ amorphous alloy after nanosecond pulsed laser processing with P_L = 10, 20 and 30 W at 0.5 μs are collected in Fig. 8. Compared with sample with P_L = 10 W, samples with P_L = 20 and 30 W have the obvious splash phe-

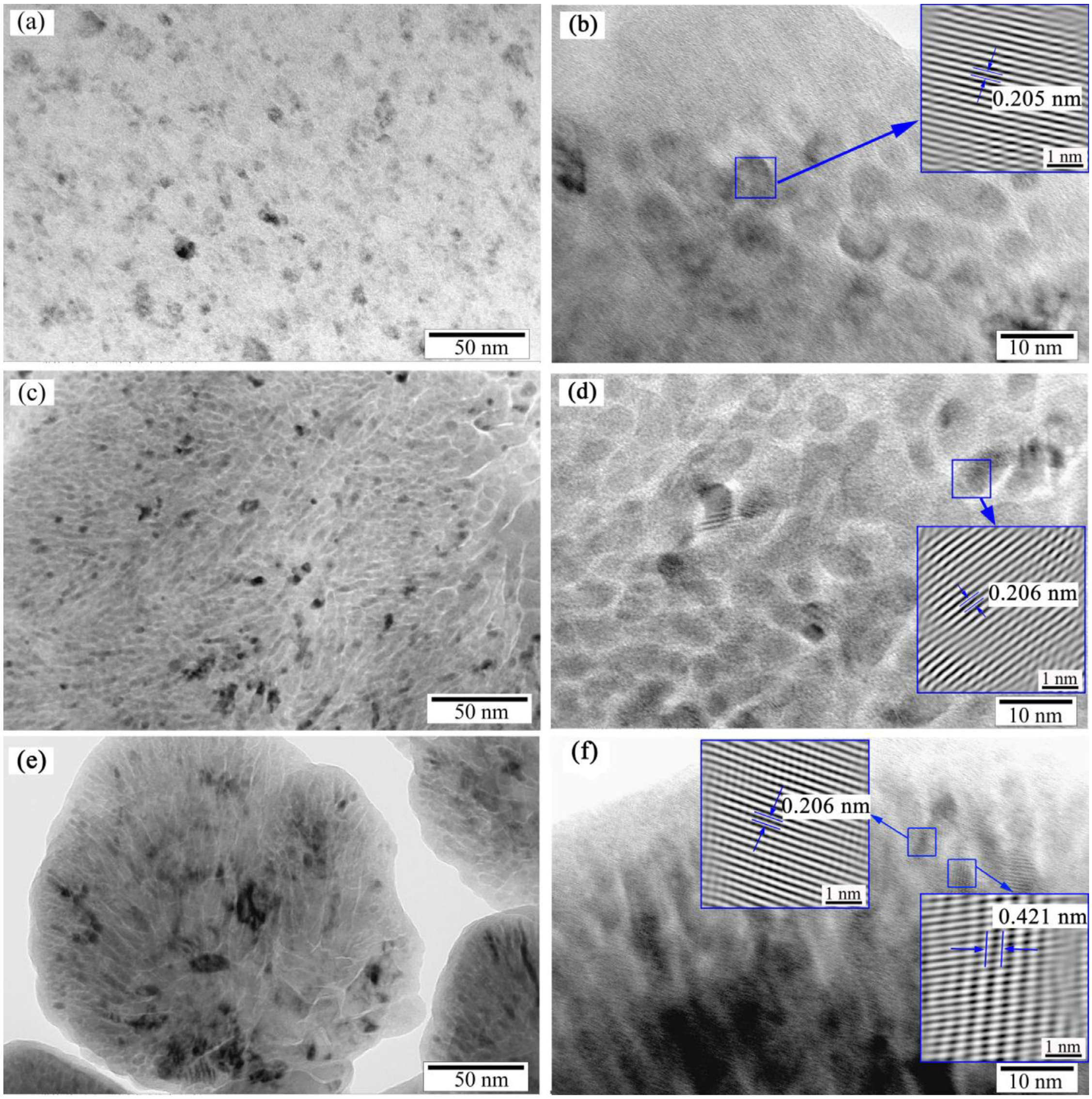

Fig. 12. TEM images of $Fe_{78}Si_9B_{13}$ amorphous alloys after nanosecond pulsed laser processing: (a) (b) for sample with P_L = 10 W, (c) (d) for sample with P_L = 20 W and (e) (f) for sample with P_L = 30 W. The insets in (b), (d) and (f) are the Fourier transformation images of selected area in HRTEM images.

nomenon (Fig. 8(a)–(c)). Based on the fluid flow field in section $Y = 0$ (Fig. 8(d)–(f)), it can be found that the v of sample with $P_L = 10$ W is relatively small and all below 4 m/s, while the maximum v of samples with $P_L = 20$ and 30 W is above 20 m/s. Moreover, the v distribution of sample with $P_L = 30$ W is more dispersed than that of sample with $P_L = 20$ W. For the pressure distribution in section $Y = 0$ (Fig. 8(g)–(i)), the pressure distribution becomes more and more chaotic with increasing P_L, which indicates that the stability of molten pool decreases. In particular, there is a large irregular low-pressure region for sample with $P_L = 30$ W, which corresponds to its severe splash phenomenon. For the temperature distribution in section $Y = 0$ (Fig. 8(j)–(l)), the surface splash phenomenon of samples with $P_L = 20$ and 30 W leads to the surface heat loss, possibly resulting in a larger surface cooling rate than sample with $P_L = 10$ W.

4.3. Structural formation analysis

Fig. 9 shows the temperature distribution of $Fe_{78}Si_9B_{13}$ amorphous alloy with different P_L and t on the top surface and cross-sectional view with $Y = 0$. At $t = 0.5$ μs, the surface temperature of the $Fe_{78}Si_9B_{13}$ amorphous alloy with $P_L = 10$, 20 and 30 W has exceeded T_l and a small molten pool has formed on the surface (Fig. 9(a)–(c)). At $t = 5$ μs, the area of molten pool area decreases with increasing P_L. For samples with $P_L = 20$ and 30 W, the splash phenomenon leads to a larger surface cooling rate and makes the surface pool area decrease significantly compared with sample with $P_L = 10$ W (Fig. 9(d)–(f)). At $t = 10$ μs, the temperature of three samples has dropped below T_l except for suspended splash droplets (Fig. 9(g)–(i)). In addition, the temperature region above T_s decreases with increasing P_L.

The temperature curves of $Fe_{78}Si_9B_{13}$ amorphous alloy during nanosecond pulsed laser processing on Z direction is shown in Fig. 10. The dotted lines in the figure show the T_l and T_c positions. Due to the flow and splash behavior of the molten pool, it is difficult to monitor the surface temperature evolution. Thus, we analyze the temperature evolution processes of sample with $P_L = 10$ W below $Z = 4$ μm and samples with $P_L = 20$ and 30 W below $Z = 6$ μm. For $P_L = 10$ W (Fig. 10(a)), the maximum temperature at 4 μm reaches T_l; the maximum temperature at $Z = 6$ and 8 μm is between T_c and T_l; the maximum temperature at $Z = 10$ μm does not exceed T_c. For $P_L = 20$ W (Fig. 10(b)), the maximum temperature at $Z = 6$ μm reaches T_l. When $Z = 8$ μm, the maximum temperature is between T_c and T_l. Below 10 μm, the maximum temperature does not exceed T_c. For $P_L = 30$ W (Fig. 10(c)), the maximum temperature reaches T_l at $Z = 6$ μm. When $Z = 8$ μm, the maximum temperature is between T_c and T_l. The maximum temperature at $Z = 10$ μm is near T_c. According to the above analysis, the range of crystallization temperature zone increases with increaing P_L. The cooling rates of samples with $P_L = 10$, 20 and 30 W from T_l to T_c at $Z = 4$, 6 and 6 μm are 2.87×10^7, 4.67×10^7 and 4.04×10^7 K/s, respectively. Here, the cooling rates of samples with $P_L = 20$ and 30 W are significantly higher than that of sample with $P_L = 10$ W. The cooling rate of sample with $P_L = 20$ W is the largest, indicating that the new amorphous phase is more easily formed for sample with $P_L = 20$ W, which is consistent with the experimental characterization [18]. In addition, the cooling rate of the region with temperature between T_l and T_c in the heat-affected zone also reaches the magnitude of 10^7 K/s, but it can be regarded as a rapid heat treatment process higher than T_c and still produces crystallization behavior. Moreover, the maximum temperature of the three samples is all above the T_c and below the T_l at $Z = 8$ μm, and the maximum temperature value increases with increasing P_L. Fig. 11 shows the time-dependent maximum depths (Z_{max}) of $Fe_{78}Si_9B_{13}$ samples with various P_L for T_l and T_c position during nanosecond pulsed laser processing. The Z_{max} of T_l and T_c for three samples increase with increasing P_L. The Z_{max} of the T_l for samples with $P_L = 10$, 20 and 30 W is 5.0, 6.5 and 7.2 μm, respectively, showing an increasing tendency with P_L. The results indicate that the molten pool area increases with increasing P_L, which is consistent with the above temperature distribution diagram (Fig. 9). The Z_{max} of T_c for samples with $P_L = 10$, 20 and 30 W is 9.2, 9.5 and 10.0 μm, respectively, indicating that the area above the T_c increases with the increasing P_L.

The TEM images of $Fe_{78}Si_9B_{13}$ amorphous alloys after nanosecond pulsed laser processing with $P_L = 10$, 20 and 30 W are given in Fig. 12. The nano-crystallites in the amorphous matrix are confirmed by the obvious dark regions in brightfield images, as shown in (Fig. 12(a), (c) and (e)). The higher P_L causes more nanocrystalline precipitation compared with low P_L. For sample with $P_L = 10$ W, the crystalline interplanar spacing in the inserted filtered lattices (Fig. 12(b)) is measured as 0.205 nm, which can be labeled as {110} of α-Fe(Si) phase. For sample with $P_L = 20$ W, the crystalline interplanar spacing in the inserted filtered lattices (Fig. 12(d)) is measured as 0.206 nm, which can be labeled as {110} of α-Fe(Si) phase. For sample with $P_L = 30$ W, the crystalline interplanar spcings are measured as 0.206 and 0.421 nm (Fig. 12(f)), which can be labeled as {110} of α-Fe(Si) phase and {002} of Fe_2B phase, respectively. The precipitated α-Fe(Si) phase in the samples with $P_L = 10$ and 20 W is only detected, while the precipitation of α-Fe(Si) and Fe-B phase could be detected in the sample with $P_L = 30$ W.

Based on the temperature analyses, the cooling rate of heat affected area is at the order of 10^7 K/s, but noncrystalline still formed owing to the flash heat treatment process above the crystallization temperature. For the $Fe_{78}Si_9B_{13}$ amorphous alloys, the crystallization usually occurs in two stages, the first one corresponding to the formation of α-Fe(Si) phase. On the subsequent stage, there is the decomposition of the remaining amorphous matrix with the formation of Fe-B phase. Based on the DSC curves in literature [18], the measured curves involve two exothermal peaks, which are labeled as Peak 1 and Peak 2, corresponding to crystallization processes of α-Fe (Si) and Fe-B, respectively. The crystallization temperature (peak temperature) for Peak 1 and Peak 2 is 799 and 820 K, respectively. According to the temperature filed analysis (Figs. 10 and 11), the peak temperature of the heat-affected area increases and the area above the crystallization temperature also increases with increasing P_L, leading to a stronger crystallization sensitivity with more nanocrystalline precipitation (Fig. 12(a), (c) and (e)). The temperature of the heat-affected area for sample with $P_L = 30$ W was higher than the samples with $P_L = 10$ and 20 W, which was more inclined to the precipitation of Fe-B phase except the α-Fe (Si) phase (Fig. 12(b), (d) and (f)). The simulations data can well explain the results of experimental characterization.

5. Conclusion

The heat transfer and melt flow coupling model for $Fe_{78}Si_9B_{13}$ amorphous alloy during nanosecond pulsed laser processing has been established in this paper. The transient distribution of flow field, temperature field and pressure field during the nanosecond pulsed laser processing and and their effects on the surface morphology and microstructure formation of $Fe_{78}Si_9B_{13}$ amorphous alloy are analyzed deeply. The main conclusions are as follows:

(1) At low P_L (10 W), the surface morphology of $Fe_{78}Si_9B_{13}$ amorphous alloy renders convex at the periphery and concave in the middle, which is similar to the coronal morphology. As P_L increase to 20 and 30 W, the surface of the sample appears obvious splash phenomenon. The simulation morphology is in good agreement with the experimental results, indicating that our numerical model is valid and can well reflect the physical processes of the $Fe_{78}Si_9B_{13}$ amorphous alloy during the nanosecond pulsed laser processing.
(2) The formation of the surface coronal-like morphology of $Fe_{78}Si_9B_{13}$ amorphous alloy after nanosecond pulsed laser processing is mainly related to the combined effect of recoil pressure, surface tension and gravity. Recoil pressure depresses the molten pool, but surface tension and gravity cushion the impact of recoil pressure. In addition, the Marangoni effect induced by the surface tension spatial gradient causes two symmetric vortexes at the molten pool edge, which plays a important role in the formation of coronal-like morphology. The velocity and pressure of melt increase with the increase of P_L, reducing the molten pool stability and triggering the splash phenomenon.
(3) Based on the temperature field, the simulated cooling rate of $Fe_{78}Si_9B_{13}$ amorphous alloy between T_l and T_c is at the order of 10^7 K/s during the nanosecond pulsed laser processing, which completely satisfies the forming conditions of Fe-based amorphous alloy. With the increase of P_L, the peak temperature of the heat-affected area increases, and the area above T_c also increases, leading to a stronger crystallization sensitivity. The simulation results are consistent well with the experimental results.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

Acknowledgment

This work was financially supported by the Key Research Project of Shandong Province (2016GGX102010) and National Natural Science Foundation of China (51771103 and 51861165202).

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